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ELECTRIC CURRENT COMPUTED TOMOGRAPHY AND EIGENVALUES*

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Abstract. In electric current computed tomography, patterns of currents or voltages are applied to electrodes on the surface of a body and the resulting voltages or currents are measured. A reconstruction and display of an approximation to the impedance inside the body is then made based on these external measurements.

It is shown how the problems of choosing current patterns, electrode size and number can be studied in terms of the spectral properties of certain pseudodifferential operators.

The proofs given here are simpler and more general than those in [*IEEE Trans. Medical Imaging*, 5 (1986), pp. 91-95] and [*Clin. Phys. Physiol. Meas. Suppl.* A, (1987), pp. 39-46].

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1. Introduction. In [2] Henderson and Webster described an "impedance camera" that applied voltages through an array of electrodes to a subject's back and chest. They measured the resulting currents, and displayed iso-admittance curves on the body surface. They then proposed the development of reconstruction algorithms for the purpose of imaging the internal impedance. Following this pioneering paper, many authors, including Webster and his co-workers, published descriptions of systems designed to reconstruct the impedance inside a body from electrical measurements made on the body's surface.

Surveys of many of these systems and their associated problems are given in Barber and Brown [3], Kim, Webster, and Tompkins [4], and the recent thesis of Yorkey [5]. Descriptions of other systems are in the patents of Bai, Brandt, and Sollish [6], Brown, Barber, and Freeston [7], and Fry and Wexler [8]. The mathematically reassuring facts that both piecewise analytic and smooth conductivities can be distinguished by boundary measurements made with infinite precision have been proven rigorously in significant papers by Kohn and Vogelius [10] and Sylvester and Uhlman [11], respectively.

An excellent study of the limits on resolution imposed by the necessarily finite precision of measurements was given in the theses of Seagar [12], [13]. A thorough literature survey, including related geophysical systems, is contained in the proceedings [9].

All of the systems referred to above operate by applying a set of patterns of current (or voltage) selected in advance by the user to an array of electrodes outside the body and recording the resulting voltages (or currents). A reconstruction algorithm is then applied to all the data accumulated and a display of the approximate internal impedance is produced.

Here we show in more detail and generality than in [1] and [14] how the problems of resolution limits, current pattern choice, and electrode size and number can be studied in terms of the spectral properties of certain pseudodifferential operators.

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2. Notation. Let B denote a body (in R^n, $n \geqq 2$) and S its surface. The "L-electrode" model consists of a subdivision of S into subsets S_l for which $S = \bigcup_{l=1}^{L} S_l$ and within each S_l there is a subset Δ_l called the electrode. Let $\Delta = |S_l| = |S|/L$, $|\Delta_l| = f\Delta$, where $\varnothing \leqq f \leqq 1$. Thus f is the fraction of surface area in each subdivision taken up by the electrode. Also let

$p =$ a point in B or on S,
$U = U(p) =$ the potential at p in B,
$V = V(p) =$ the voltage at p on S,
$j = j(p) =$ the inward normal component of current density on S,
$\sigma = \sigma(p) =$ the conductivity at p in B,

$$X_{\Delta_j}(p) = \begin{cases} 1 & \text{for } p \in \Delta_j \\ 0 & \text{for } p \notin \Delta_j \end{cases}.$$

If $F(p)$ and $G(p)$ are functions on S then define

$$\langle F, G \rangle = \int_S F(p) G(p)\, dA,$$

$$\|F\| = \left[\int_S |F(p)|^2\, dA \right]^{1/2}$$

$$= [\langle F, F \rangle]^{1/2}.$$

3. Forward problem. In the forward problem we assume the conductivity σ is known and that we can apply current densities j_l; $l = 1, 2, \cdots$ to S and measure voltages V_l on S where

$$\nabla \cdot \sigma \nabla U = 0 \quad \text{in } B,$$

$$\sigma \frac{\partial u}{\partial n} = j_l \quad \text{on } S.$$

For existence and uniqueness we require j_l and $V_l(p) \equiv U(p)$ when $p \in S$, to satisfy

$$\int_S j_l\, dA = \int_S V_l(p)\, dA = 0.$$

If $\sigma \in L^\infty(B)$ and $\sigma \geqq \sigma_0 > 0$ then there exists a unique $V_l \in H^{1/2}(S)$† for each $j_l \in H^{-1/2}(S)$. We denote the linear operator that assigns V_l to j_l by $R(\sigma)$. Thus $R(\sigma)$: $H^{-1/2}(S) \to H^{1/2}(S)$

$$R(\sigma) j_l = V_l.$$

When σ is C^∞ in a neighborhood of S, $R(\sigma)$ is a pseudodifferential operator, the inverse of the "Calderon map" [15], [16], [10], [11].

Properties of $R(\sigma)$:

(1) Depends nonlinearly on σ.
(2) Compact and self-adjoint as a map from $H^0(S) \to H^0(S)$.
(3) $\sigma_1 \geqq \sigma_2 \to R(\sigma_1) \leqq R(\sigma_2)$ as forms on $H^0(S) \times H^0(S)$.
(4) $R(\sigma)$ is determined by the "spectral data" $\{\lambda_k(\sigma), \phi_k(p; \sigma)\}$ for the operator $L = -\nabla \cdot \sigma \nabla$ in B with Neumann boundary conditions on S. Here λ_k and ϕ_k for

† We write $H^p(S)$ for the subspace of the usual H^p that is orthogonal to the constants.

$k=1,2,\cdots$ denote the eigenvalues and boundary values of the eigenfunctions $\Phi_k = \Phi_k(p;\sigma)$ where $\lambda_1 \leqq \lambda_2 \leqq \cdots$

$$L\Phi_k = \lambda_k \Phi_k \quad \text{in } B,$$
$$\partial_n \Phi_k = 0 \quad \text{on } S,$$
$$\int_B |\Phi_k(p)|^2 \, d_p = 1,$$

and

$$\phi_k(p;\sigma) \equiv \Phi_k(p,\sigma) \quad \text{for } p \quad \text{on } S.$$

The proofs of (3) and (4) are outlined in Appendices I and II, respectively.

We remark that in practice we cannot arbitrarily specify the current densities j_l but only the currents. The measured voltages are also *not* exactly linear functions of the currents because of measurement and model errors which we do not go into here. In what follows $\tau = \tau(p)$ will also denote a possible conductivity throughout B.

4. Problem 1. Can we distinguish σ from τ by measurements of finite precision?

In order to give an experimental procedure with which to answer this question we introduce a definition of distinguishability [1].

DEFINITION. Conductivities σ and τ are distinguishable, in the mean square sense, by measurements of precision ε if there is a current density $j \in H^0$ for which

$$\delta = \delta(j) = \frac{\|R(\sigma)j - R(\tau)j\|}{\|j\|} > \varepsilon.$$

The conductivities σ and τ are not distinguishable by measurements of precision ε if for all $j \in H^0$, $\delta(j) \leqq \varepsilon$. We call the number $\delta(j)$ the "distinguishability." It is a function of σ, τ, and j.

Remark. More general notions are needed for all j's with finite energy, $j \in H^{-1/2}(S)$. In particular, we use

$$\delta^{(q)}(j) = \frac{\|R(\sigma)j - R(\tau)j\|_{+q}}{\|j\|_{-q}},$$

where

$$\|F\|_q^2 = \int_S |(1-\Delta)^{q/2} F(p)|^2 \, dA.$$

Here Δ denotes the Laplace–Beltrami operator on S and $q=\frac{1}{2}$. Unless σ and τ differ on the boundary S the results obtained with $q=0$ are similar to those with $q=\frac{1}{2}$.

DEFINITION. The "best" currents $\tilde{j}$ to distinguish σ from τ, in the mean square sense, are those that maximize $\delta(j)$:

$$\delta(\tilde{j}) = \max_{j \in H^0} \delta(j)$$
$$= \max \frac{\|[R(\sigma) - R(\tau)]j\|}{\|j\|}.$$

Thus

$$\delta(\tilde{j})^2 = \max_{\|j\|=1} \langle j, D^2 j \rangle,$$

where

$$D = |R(\sigma) - R(\tau)|.$$

The relevant properties of D, which maps H^0 to H^1, are:

(1) D is a compact, self-adjoint, pseudodifferential operator on $H^0(S)$ to $H^0(S)$.

(2) D has a complete orthonormal set of eigenfunctions $\psi_k(p) \in C^\infty(S)$, $k = 1, 2, \cdots$ whose eigenvalues $\delta_1 \geqq \delta_2 \geqq \cdots$ satisfy $\lim_{k\to\infty} \delta_k = 0$. If σ and τ agree in a neighborhood of S, $\delta_k = O(e^{-ck^\alpha})$ where c depends on the size of the neighborhood and α depends on the dimension. See Appendix III for a simple example illustrating this exponential decay. Thus

$$D\psi_k = \delta_k \psi_k,$$

$$\langle \psi_\alpha, \psi_\beta \rangle = \begin{Bmatrix} 0 & \text{if } \alpha \neq \beta \\ 1 & \text{if } \alpha = \beta \end{Bmatrix}$$

and for any $j \in H^0(S)$

$$j = \sum_{k=1}^{\infty} \langle j, \psi_k \rangle \psi_k.$$

It follows from the min-max principle [17] that the largest value the distinguishability can have is δ_1 where δ_1 is D's largest eigenvalue. The "best" currents are therefore the eigenfunctions $\psi_1, \cdots$ of D whose eigenvalue is δ_1 [1].

5. Problem 2. What happens to the distinguishability of a single current applied between a pair of electrodes (Δ_1 and Δ_l) as the electrode size $\Delta \to 0$, and $L \to \infty$?

The solution to this problem is the theorem

$$\lim_{\substack{L\to\infty \\ \Delta\to 0}} \delta(j_{\text{pair}}) = 0.$$

In other words, as we increase the number of electrodes to infinity in order to obtain higher resolution reconstructions, the signal-to-noise ratio goes to zero for single currents applied on a pair of electrodes.

Proof. By a single current applied on the pair of electrodes Δ_1 and Δ_l we mean any current density $j_{\text{pair}} \in H^0(S)$ where

$$j_{\text{pair}} = j_{\text{pair}}[X_{\Delta_1} + X_{\Delta_l}].$$

Now

$$j_{\text{pair}} = \sum_{k=1}^{\infty} \langle j_{\text{pair}}, \psi_k \rangle \psi_k,$$

$$Dj_{\text{pair}} = \sum_{k=1}^{\infty} \langle j_{\text{pair}}, \psi_k \rangle \delta_k \psi_k,$$

$$\delta(j_{\text{pair}})^2 = \frac{\|Dj_{\text{pair}}\|^2}{\|j_{\text{pair}}\|^2} = \frac{\sum \delta_k^2 \langle j_{\text{pair}}, \psi_k \rangle^2}{\sum \langle j_{\text{pair}}, \psi_k \rangle^2}.$$

For any $\varepsilon > 0$, choose K so large that $\delta_{K+1} < \varepsilon/2$. Then

$$\delta^2 \leqq \frac{\sum_{k=1}^{K} \delta_k^2 \langle j_{\text{pair}}, \psi_k \rangle^2}{\|j_{\text{pair}}\|^2} + \delta_{K+1}^2.$$

However,

$$|\langle j_{\text{pair}}, \psi_k\rangle| = |\langle (X_{\Delta_l} + X_{\Delta_1}) j_{\text{pair}}, \psi_k\rangle| < \|j_{\text{pair}}\| \, \|(X_{\Delta_1} + X_{\Delta_l})\psi_k\|.$$

Since

$$\|(X_{\Delta_1} + X_{\Delta_l})\psi_k\| \leqq 2\sqrt{f\Delta} \max_{p \in S} |\psi_k(p)| \leqq 2\sqrt{f\Delta}\, M_k,$$

$$\lim_{\Delta \to 0} \|(X_{\Delta_1} + X_{\Delta_l})\psi_k\| = 0.$$

It follows that

$$\delta(j_{\text{pair}})^2 \leqq K 4 f \Delta \delta_1^2 \max_{1 \leqq k \leqq K} M_k + \frac{\varepsilon}{2}$$

and since ε was arbitrary

$$\lim_{\Delta \to 0} \delta(j_{\text{pair}}) = 0.$$

We remark that $\delta \approx (\text{const.}) \sqrt{f\Delta}$ as $\Delta \to 0$.

If σ and τ differ *on* S then for some $j_{\text{pair}} \in H^{-1/2}(S)$, $\delta^{(1/2)}(j_{\text{pair}})$ may not approach zero. However if σ and τ are equal in a neighborhood of S again we have $\delta^{(1/2)}(j_{\text{pair}}) \to 0$ as $\Delta \to 0$. We do not give this technically more difficult proof here.

6. Problem 3. Here we define approximations j_L to the best current ψ_1, on L-electrode systems, and ask what happens to the distinguishability of j_L as $L \to \infty$?

DEFINITION. J_L is an "L-electrode approximation" to the best current density ψ_1 if and only if

$$J_L = \frac{P_L \psi_1}{\|P_L \psi_1\|} + \rho_L,$$

where

$$P_L = \sum_{l=1}^{L} X_{\Delta_l}$$

and

$$\|\rho_L\| \to 0 \quad \text{as } L \to \infty.$$

As an example we might take

$$\bar{j}_l \equiv \frac{1}{|\Delta|} \int_{S_l} \psi_1(p)\, dA$$

and

$$j_L(p) = \frac{\sum_{l=1}^{L} \bar{j}_l X_{\Delta_l}(p)}{\|\sum_{l=1}^{L} \bar{j}_l X_{\Delta_l}(p)\|}.$$

We may now give the solution to Problem 3 as the following theorem.

THEOREM.

$$\lim_{\substack{\Delta \to 0 \\ L \to \infty}} \delta(j_L) = \sqrt{f}\, \delta_1.$$

This theorem says that "L-electrode" approximations to the best currents have distinguishability that approaches $\sqrt{f}$ times its maximum possible value as the number of electrodes goes to infinity, suggesting that to maximize the signal-to-noise ratio we should use electrodes that are as large as possible ($f \approx 1$) and approximations to the best current densities ψ_1. The intuitive reason why the electrodes should be as large as possible is that the "best" current densities are smooth functions, and a smooth function is better approximated by L-electrode approximations with "small gaps."

Before describing an adaptive experimental method for producing approximations to ψ_1, we give the proof.

Proof.

$$\delta(j_L)^2 = \sum_{k=1}^{\infty} \delta_k^2 \langle j_L, \psi_k \rangle^2,$$

$$\langle j_L, \psi_k \rangle = \left\langle \frac{P_L \psi_1}{\|P_L \psi_1\|}, \psi_k \right\rangle + O(\|\rho_L\|).$$

The ψ_k are in $C^\infty(S)$ so that

$$\begin{aligned}\langle P_L \psi_1, \psi_k \rangle &= \sum_{l=1}^{L} \int_{\Delta_l} \psi_1(p) \psi_k(p)\, dA \\ &= \sum_{l=1}^{L} \psi_1(\xi_l) \psi_k(\xi_l) f \Delta,\end{aligned}$$

where $\xi_l \in \Delta_l$. Thus as $\Delta \to 0$ and $L \to \infty$

$$\langle P_L \psi_1, \psi_k \rangle \to f \langle \psi_1, \psi_k \rangle = f \delta_{1,k}.$$

Similarly,

$$\|P_L \psi_1\|^2 = \langle P_L \psi_1, \psi_1 \rangle \to f$$

so

$$\langle j_L, \psi_k \rangle \to \left\{ \begin{matrix} \sqrt{f} & \text{if } k = 1, \text{ and} \\ 0 & \text{if } k > 1 \end{matrix} \right\}.$$

Hence by an argument similar to that in Problem 2,

$$\lim_{\substack{L \to \infty \\ \Delta \to 0}} \delta(j_L) = \sqrt{f}\, \delta_1.$$

We are now ready for the main problem.

7. Problem 4. How can we produce approximations, accurate to within the measurement precision of the system in use, to the best current densities ψ_1, and the maximum distinguishability δ_1, without any knowledge of σ?

Once we have such approximations we can decide if our guess τ for the conductivity is in fact distinguishable from σ. If it is, we may improve it. If it is not, there is no point in changing τ because it is within our measurement ability indistinguishable from σ.

We now describe a simple experimental process for finding ψ_1 and δ_1.

1. Guess any current density $j^{(0)}$ for which $\|j^{(0)}\| = 1$ and $\int_S j^{(0)}\, dA = 0$. Set $k = 0$.
2. *Measure*

$$V_\sigma^{(k)} = R(\sigma) j^{(k)}.$$

3. *Compute*

$$V_\tau^{(k)} = R(\tau) j^{(k)}.$$

4. *Compute approximations*

$$\delta_1^{(k)} = \| V_\sigma^{(k)} - V_\tau^{(k)} \|$$

$$j^{(k+1)} = \frac{V_\sigma^{(k)} - V_\tau^{(k)}}{\delta_1^{(k)}}.$$

5. Test if $\| j^{(k+1)} - j^{(k)} \|$ is less than the measurement precision. If it is stop and use $j^{(k)} \approx \psi_1$ and $\delta_1^{(k)} \approx \delta_1$. If it is greater than the measurement precision set $k = k+1$ and go to 2.

THEOREM. *If* $\delta_1 > \delta_2$ *and* $\langle j^{(0)}, \psi_1 \rangle \neq 0$, *then* $\lim_{k \to \infty} j^{(k)} = \pm \psi_1$ *and* $\lim_{k \to \infty} \delta^{(k)} = \delta_1$.

The proof of this theorem and the fact that the convergence is $O((\delta_2/\delta_1)^n)$, i.e., exponentially fast, is essentially the proof of convergence of the "power method" for finding the largest eigenvalue of a matrix (see, for example, [18]).

8. Problem 5. The inverse problem is that of trying to reconstruct an approximation τ, in some class of functions Σ, to the actual conductivity σ in the body. In this section we give a heuristic estimate for the number of degrees of freedom N in the set Σ and the number of electrodes L that are appropriate for a system with measurements of precision ε. Here ε is the smallest value of the distinguishability justified by the precision of the voltmeters and current generators used.

Note first that if measurements are made with perfect precision, $\varepsilon = 0$, on an L electrode system there are at most $L(L-1)/2$ independent equations possible because there are only $L-1$ independent currents that can be applied, and the operator $R(\sigma)$ is self-adjoint. Thus the number of degrees of freedom in Σ that could be uniquely determined from these equations must be less than or equal to $L(L-1)/2$, i.e.,

$$N \leqq L(L-1)/2.$$

When $\varepsilon = 0$, $L = \infty$ is appropriate. When $\varepsilon > 0$ we estimate L in the simple case in which B is the unit disk, $x^2 + y^2 \leqq 1$, and Σ is a class of functions defined on a mesh of J radial and K angular subdivisions of B. For example, $\tau \in \Sigma$ could be given by $\tau(p) = \sum_{n=1}^N \tau_n \Phi_n(p)$, where $\Phi_n(p)$ is a finite-element basis function that is 1 on the nth node in the mesh and zero at the other nodes.

The number of degrees of freedom is then

$$J \cdot K = N.$$

Choose the size of the smallest radial subdivision equal to the size of the smallest circular inhomogeneity that can be distinguished from a uniform background by measurements of precision ε [1], [12], [13]. For example, if the background has conductivity 1 and the inhomogeneity has conductivity σ and radius r, then $r \approx (\varepsilon/2|\mu|)^{1/2}$ where $\mu = (\sigma-1)/(\sigma+1)$. Thus $J = 1/r$. Choose the number of angular subdivisions K equal to the number of electrodes, i.e., $K = L$:

$$\frac{1}{r} \cdot L = N \leqq L(L-1)/2,$$

which suggests that

$$L \geqq \frac{2}{r} + 1 \approx 2(2|\mu|/\varepsilon)^{1/2} + 1.$$

Thus the smallest number of electrodes for which the number of degrees of freedom equals the maximum possible number of independent equations is

$$L \approx 2(2|\mu|/\varepsilon)^{1/2}+1.$$

We do not give here the more detailed analysis necessary to show that the use of significantly fewer electrodes than this would result in a loss of resolution and the use of significantly more electrodes would result in little gain in resolution near the center and a more highly ill-conditioned inverse problem. We merely remark that using a mesh (or class of functions Σ) that is significantly finer than the smallest variations in conductivity that the system can distinguish will result in greater ill-conditioning in whatever scheme is used to reconstruct σ.

Appendix I. Monotonicity. We prove that the energy dissipated in heat by a body subjected to an external current distribution increases as the body's internal resistance increases. In other words,

$$\sigma(p) \geqq \tau(p) \quad \text{for all } p \text{ in } B$$

implies

$$R(\sigma) \leqq R(\tau) \quad \text{as forms on } H^0(S)\times H^0(S).$$

Proof. Let

$$P_\sigma(\omega)=\frac{1}{2}\int_B \sigma|\nabla\omega|^2\,dp-\int_S \omega j\,dA.$$

If ω is a solution to

$$\nabla\cdot\sigma\nabla\omega=0 \quad \text{in } B,$$

$$\sigma\frac{\partial\omega}{\partial n}=j \quad \text{on } S.$$

Then for any smooth function $\eta(p)$ in B

$$\text{(A1)}\qquad \begin{aligned} &\int_B \sigma|\nabla\omega|^2\,dp=\int_S j\omega\,dA=\langle j, R(\sigma)j\rangle,\\ &\int_B \sigma\nabla\omega\cdot\nabla\eta\,dp=\int_S \eta j\,dA. \end{aligned}$$

These yield the inequality [17]

$$\text{(A2)}\qquad \begin{aligned} P_\sigma(\omega+\eta)&=P_\sigma(\omega)+\frac{1}{2}\int_B \sigma|\nabla\eta|^2\,dp\\ &\geqq P_\sigma(\omega). \end{aligned}$$

We now show that $\langle j, R(\sigma)j\rangle \leqq \langle j, R(\tau)j\rangle$ when $\sigma\geqq\tau$ in B. Let $U_\sigma=R(\sigma)j$ and $U_\tau=R(\tau)j$. From (A1)

$$\text{(A3)}\qquad \langle j, U_\sigma\rangle=\int_B \sigma|\nabla U_\sigma|^2\,dp,\qquad \langle j, U_\tau\rangle=\int_B \tau|\nabla U_\tau|^2\,dp,$$

where we use the same notation U_* for a function in B and its boundary value on S. For an arbitrary smooth function U on B

$$P_\sigma(U) \geqq P_\tau(U)$$

because $\sigma \geqq \tau$. From (A2)

$$P_\tau(U) \geqq P_\tau(U_\tau).$$

Thus taking $U = U_\sigma$ we have

$$P_\sigma(U_\sigma) \geqq P_\tau(U_\tau).$$

From the definition of P_σ and the identities (A3) it follows that

$$\tfrac{1}{2}\langle j, U_\sigma\rangle - \langle U_\sigma, j\rangle \geqq \tfrac{1}{2}\langle j, U_\tau\rangle - \langle U_\tau, j\rangle,$$

which is equivalent to

$$\langle R(\sigma)j, j\rangle \leqq \langle R(\tau)j, j\rangle.$$

Appendix II. Connections between inverse boundary value and inverse spectral problems. We prove the formula

$$[R(\sigma)j](\xi) = \int_S g(x, \xi)j(x)\, dS_x, \tag{$*$}$$

where

$$g(x, \xi) = \sum_{k=1}^{\infty} \frac{1}{\lambda_k} \phi_k(x)\tilde{\phi}_k(\xi),$$

$$\tilde{\phi}_k(\xi) = \phi_k(\xi) - \frac{1}{|S|}\int_S \phi_k(\xi)\, dS_\xi,$$

and for x on S

$$\phi_k(x) = \Phi_k(x).$$

Here $0 = \lambda_0 < \lambda_1 \leqq \lambda_2 \leqq \cdots$, and Φ_k denote the eigenvalues and eigenfunctions of $L = -\nabla \cdot \sigma\nabla$, i.e.,

$$L\Phi_k = -\nabla \cdot \sigma\nabla\Phi_k = \lambda_k\Phi_k \quad \text{in } B,$$

$$\partial_n\Phi_k = 0 \quad \text{on } S,$$

$$\int_B \Phi_k(x)^2\, dx = 1.$$

Formula ($*$) implies that the boundary operator $R(\sigma)$ is completely determined by the spectral data $M(\sigma) \equiv \{\lambda_k(\sigma), \phi_k(p; \sigma)\}_{k=1}^{\infty}$.

An inverse boundary value problem in electrical tomography is to determine σ from $R(\sigma)$. An inverse spectral problem is to determine σ from $M(\sigma)$. Formula ($*$) says that $R(\sigma_1) \neq R(\sigma_2) \Rightarrow M(\sigma_1) \neq M(\sigma_2)$. Thus uniqueness theorems for the inverse boundary value problem imply uniqueness theorems for the inverse spectral problem. In particular, Kohn and Vogelius [10] have proven that if σ_1 and σ_2 are piecewise analytic and $\sigma_1 \neq \sigma_2$ (in dimensions greater than one), then $R(\sigma_1) \neq R(\sigma_2)$. Sylvester and Uhlman [11] have proven that (in dimensions greater than two) if σ_1 and σ_2 are C^∞ and $\sigma_1 \neq \sigma_2$, then $R(\sigma_1) \neq R(\sigma_2)$. These results and ($*$) imply that $M(\sigma_1) \neq M(\sigma_2)$ for piecewise analytic or C^∞ functions $\sigma_1 \neq \sigma_2$ in dimensions greater than one and two, respectively.

Formula $(*)$ also yields an interesting relation between the spectral representations of $R(\sigma)$ and L. If we denote the eigenvalues and eigenfunctions of $R(\sigma)$ by ρ_n and $\psi_n(x)$, i.e., $R(\sigma)\psi_n=\rho_n\psi_n$, then from $(*)$

$$\rho_n\psi_n(\xi)=[R(\sigma)\psi_n](\xi)=\sum_{k=1}^{\infty}\frac{1}{\lambda_k}\tilde{\phi}_k(\xi)\langle\phi_k,\psi_n\rangle$$

and

$$\rho_n=\sum_{k=1}^{\infty}\frac{1}{\lambda_k}[\langle\phi_k,\psi_n\rangle]^2.$$

We give the proof of $(*)$.

Proof. Note that $\Phi_0(x)=\text{const.}=|B|^{-1/2}$. From the completeness of the eigenfunctions of L we have that

$$\sum_{k=0}^{\infty}\Phi_k(x)\Phi_k(\xi)=\delta(x-\xi).$$

Let

$$G(x,\xi)\equiv\sum_{k=1}^{\infty}\frac{1}{\lambda_k}\Phi_k(x)\Phi_k(\xi).$$

Then $L_xG(x,\xi)=\sum_{k=1}^{\infty}\Phi_k(x)\Phi_k(\xi)=\delta(x-\xi)-|B|^{-1}$ and $\partial_{n_x}G(x,\xi)=0$ for $x\in S$. Thus if $L_x\tilde{u}(x)=0$, $\sigma\partial_{n_x}\tilde{u}(x)=j(x)$, and $\int_B\tilde{u}(x)\,dx=0$, then

$$\begin{aligned}\tilde{u}(\xi)&=\int_B\tilde{u}(x)L_xG(x,\xi)-G(x,\xi)L_x\tilde{u}(x)\,dx\\&=\int_S G(x,\xi)j(x)\,dS_x.\end{aligned}$$

Now for $\xi\in S$, $u(\xi)=[R(\sigma)j](\xi)=\tilde{u}(\xi)+\delta u$ where δu is a constant. Since

$$0=\int_S u(\xi)\,dS_\xi=\int_S\tilde{u}(\xi)\,dS_\xi+|S|\delta u,$$

we have that

$$\delta u=-\frac{1}{|S|}\int_S\tilde{u}(\xi)\,dS_\xi,$$

and therefore when $\xi\in S$

$$\begin{aligned}u(\xi)=[R(\sigma)j](\xi)&=\tilde{u}(\xi)-\frac{1}{|S|}\int_S\tilde{u}(\xi)\,dS_\xi\\&=\int_S\left\{G(x,\xi)-\frac{1}{|S|}\int_S G(x,\xi)\,dS_\xi\right\}j(x)\,dS_x\\&=\int_S g(x,\xi)j(x)\,dS_x,\end{aligned}$$

where

$$g(x,\xi)=\sum_{k=1}^{\infty}\frac{1}{\lambda_k}\phi_k(x)\tilde{\phi}_k(\xi).$$

Appendix III. Bounding the eigenvalues of D. We sketch a simple proof of the exponential decay of the eigenvalues of $D=|R(\sigma)-R(\tau)|$ in the special case in which B is the disk of radius one, i.e.,

$$B=\{p=(x,y)\,|\,|p|^2=x^2+y^2\leqq 1\}$$

and

$$\sigma(p)=\tau(p)=1 \quad \text{for } r\leqq|p|\leqq 1.$$

Let

$$m_\sigma=\min_B \sigma(p), \qquad M_\sigma=\max_B \sigma(p),$$

$$\sigma_m=\sigma_m(p)=\left\{\begin{matrix} m_\sigma & \text{for } |p|\leqq r \\ 1 & \text{for } r<|p|\leqq 1 \end{matrix}\right\},$$

$$\sigma_M=\sigma_M(p)=\left\{\begin{matrix} M_\sigma & \text{for } |p|\leqq r \\ 1 & \text{for } r<|p|\leqq 1 \end{matrix}\right\}.$$

Similarly define m_τ, M_τ, $\tau_m(p)$, and $\tau_M(p)$.

From the monotonicity result in Appendix I and the fact that

$$\sigma_m(p)\leqq\sigma(p)\leqq\sigma_M(p), \qquad \tau_m(p)\leqq\tau(p)\leqq\tau_M(p),$$

it follows that

$$R(\sigma_M)-R(\tau_m)\leqq R(\sigma)-R(\tau)\leqq R(\sigma_m)-R(\tau_M).$$

This shows that the eigenvalues of $R(\sigma)-R(\tau)$ are squeezed between those of $R(\sigma_M)-R(\tau_m)$ and $R(\sigma_m)-R(\tau_M)$. These eigenvalues are explicitly given in [1] by

$$\frac{1}{k}\left[\frac{1-\mu_a r^{2k}}{1+\mu_a r^{2k}}-\frac{1-\mu_b r^{2k}}{1+\mu_b r^{2k}}\right]=O\left[\frac{1}{k}r^{2k}\right],$$

where $\mu_z=(z-1)/(z+1)$, $a=m_\sigma$, $b=M_\tau$ for the upper bound, and $a=M_\sigma$, $b=m_\tau$ for the lower bound.

Thus the $\delta_k=O[e^{-ck}]$ where $C=-2\ln r$.

We remark that it can also be shown that for large k the eigenfunctions $\psi_k(p)$ approach those of $R(1)$, i.e., $\cos k\theta$ and $\sin k\theta$ in this simple case.

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Note added in proof. Since the acceptance of this paper in 1987, many deeper studies of this and closely related boundary value problems have appeared.

We especially refer the interested reader to : A. Nachman, J. Sylvester, and G. Uhlmann, *An n-dimensional Borg–Levinson theorem*, Comm. Math. Phys., 115 (1988), pp. 595–605; A. Nachman, *Reconstructions from boundary measurements*, Ann. of Math., 128 (1988), pp. 531–587; and the excellent survey paper by J. Sylvester and G. Uhlmann, *The Dirichlet to Neumann map*, in Inverse Problems in Partial Differential Equations, D. Colton, R. Ewing, and W. Rundell, eds., Society for Industrial and Applied Mathematics, Philadelphia, 1990, pp. 101–139.

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